

COMPUTER VISION – DATA INTERPRETATION & ASSESSMENT

Course Assessment Tool (CO4_AT1) | Technical Evaluation & Case Solutions

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Question 1: Real-Time Detection System Selection

Method	Accuracy (%)	Speed / Latency Profile
Viola-Jones	78%	Fast (< 20 ms)
YOLO (Selected)	92%	Moderate (~30–45 ms)
Deep Learning (Heavy)	95%	Slow (> 100 ms)

Evaluation & Comparative Analysis

- 1. Viola-Jones (78% Accuracy, Fast Speed):** Relies on Haar-like feature cascades and sliding window detection. While computationally lightweight and fast, its low accuracy (78%) leads to excessive false negatives and misclassifications in dynamic outdoor environments.
- 2. Heavy Deep Learning Models (95% Accuracy, Slow Speed):** Achieves superior accuracy (95%) using deep multi-stage architectures (e.g., Faster R-CNN). However, high inference latency (>100 ms per frame) drops frame rates below 10 FPS, causing frame drops and failing strict latency constraints.
- 3. YOLO - You Only Look Once (92% Accuracy, Moderate Speed):** Frames object detection as a single unified regression problem. It processes full images in a single network pass, achieving 92% accuracy at ~30–45 ms latency (~25-30 FPS).
- Key Recommendation / Verdict:** YOLO should be selected for real-time surveillance. It strikes the optimal balance on the Pareto frontier—offering high detection accuracy (92%) while strictly satisfying frame-rate constraints (<50 ms latency).

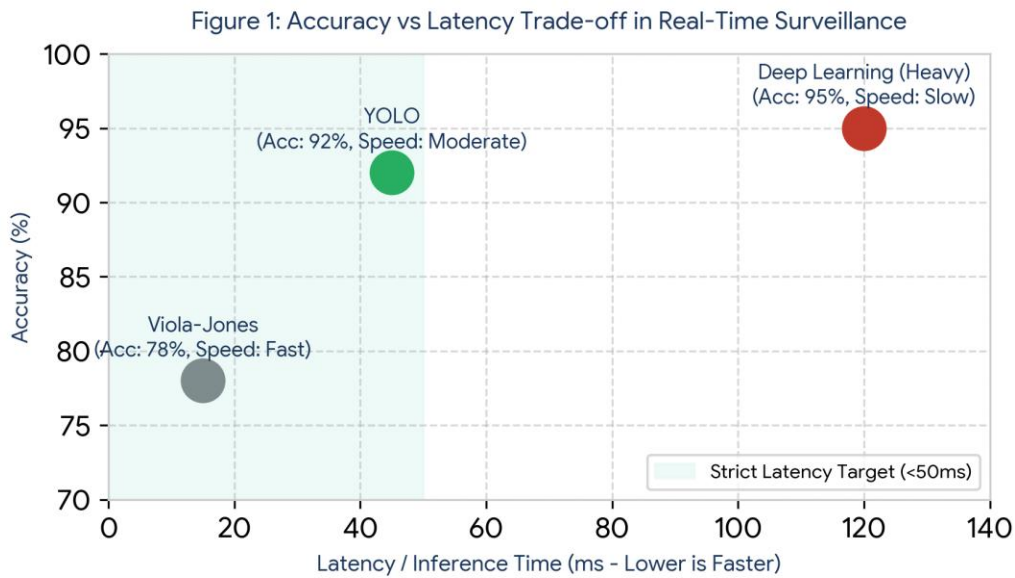


Figure 1: Accuracy vs Latency trade-off curve across candidate algorithms.

Question 2: Robust Object Detection Under Environmental Variations

Method	Normal Acc (%)	Occlusion Acc (%)	Low Light Acc (%)
Viola-Jones	80%	60%	55%
YOLO (Best Overall)	90%	75%	70%
Heavy Deep Learning	95%	85%	80%

Robustness Evaluation & Trade-off Discussion

- Performance Degradation:** Viola-Jones suffers severe degradation under low light (55%) and occlusion (60%) because simple edge features fail when contrast drops. Heavy Deep Learning maintains higher resilience (80-85%) due to deep hierarchical representations.
- Trade-off Evaluation:** While Heavy Deep Learning achieves slightly higher accuracy under adverse conditions, it requires prohibitively high GPU hardware costs and causes processing latency. YOLO maintains strong robust performance (75% occlusion, 70% low light) while remaining edge-deployable.

Key Recommendation / Verdict: YOLO provides the best overall performance, maintaining reliable feature extraction in low-light and occluded settings while preserving real-time execution speeds.

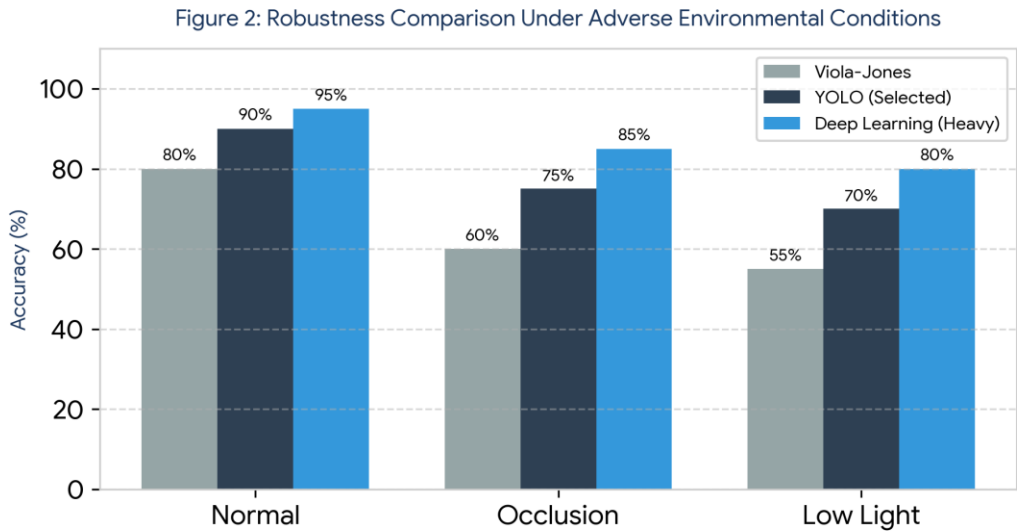


Figure 2: Robustness comparison of algorithms across Normal, Occluded, and Low Light conditions.

Question 3: Optical Flow Analysis for Motion Tracking

Method	Motion Stability	Noise Sensitivity	Fine Motion Detection	Recommended Application
Method A (Lucas-Kanade / Smooth)	High (Smooth Vectors)	Low (High Noise Immunity)	Low (Misses micro-motion)	Dynamic Motion Tracking
Method B (Horn-Schunck / Dense)	Low (Noisy Vectors)	High (Sensitive to Noise)	High (Detects subtle motion)	Micro-Motion / Texture Analysis

Evaluation for Dynamic Tracking Scenes

In dynamic tracking scenarios (e.g., surveillance or vehicle tracking), motion estimation algorithms face sensor noise, camera jitter, and changing illumination. **Method A is superior for tracking** because stable, noise-immune motion vectors prevent false track trajectories. Method B generates noisy vectors that lead to false motion estimates and broken target trajectories.

Key Recommendation / Verdict: Method A is selected for motion tracking in dynamic scenes. Vector stability and noise robustness are essential to maintain accurate, continuous object trajectories.

Figure 3: Optical Flow Motion Vector Comparison (Method A vs Method B)

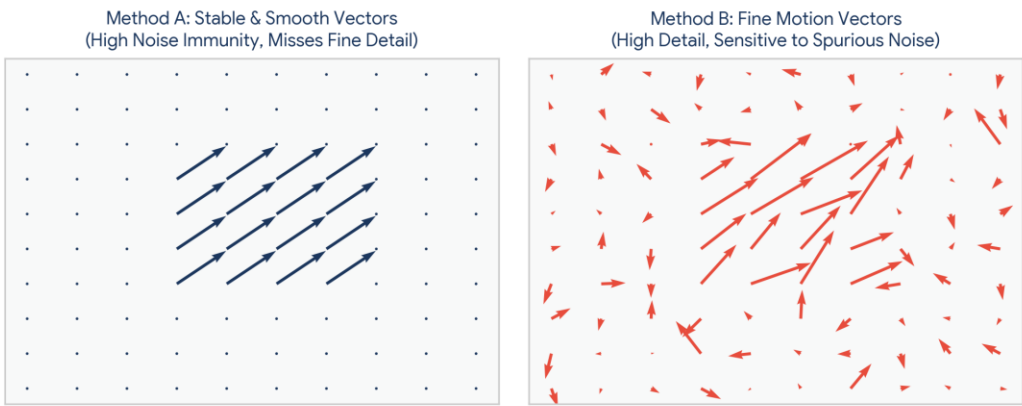


Figure 3: Comparison of optical flow vector fields generated by Method A (Stable) and Method B (Noisy).

Question 4: Model Generalization and Overfitting Analysis

Model	Training Accuracy	Testing Accuracy	Generalization Gap
Model A	98%	70%	28% (Severe Overfitting)
Model B (Selected)	90%	88%	2% (High Generalization)

Generalization Ability & Overfitting Risk

- Model A (Severe Overfitting):** Model A exhibits a large 28% gap between training (98%) and testing (70%) accuracy. This indicates that Model A has memorized noise and specific training set details, making it highly unreliable for real-world deployment.
- Model B (High Generalization):** Model B maintains a narrow 2% gap between training (90%) and testing (88%) accuracy. It has learned robust generalized feature representations, ensuring predictable performance on unseen test data.

Key Recommendation / Verdict: Model B is selected for real-world deployment. Its minimal generalization gap ensures high operational reliability and low vulnerability to out-of-distribution inputs.

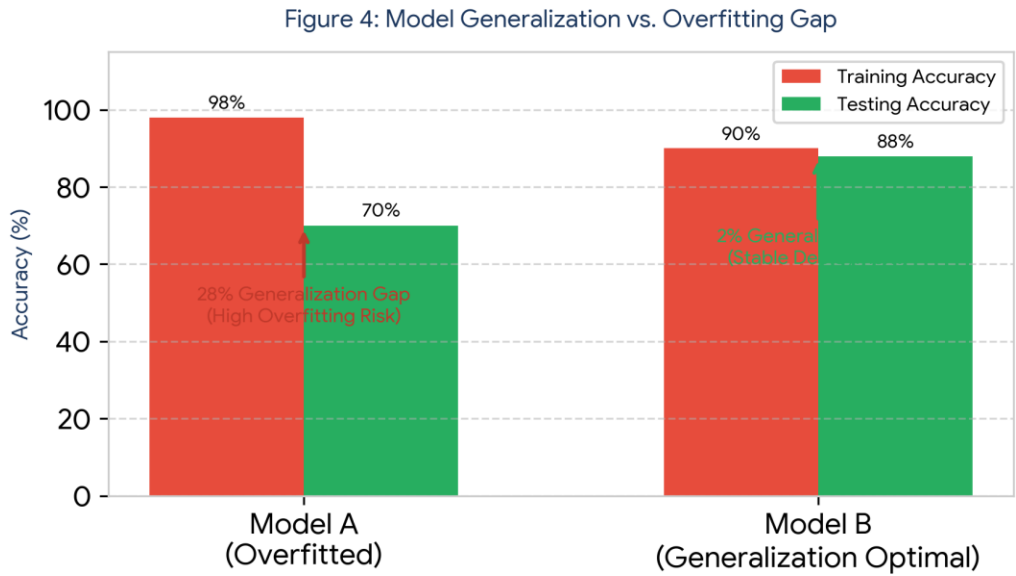


Figure 4: Generalization gap between training and testing accuracy for Model A and Model B.

Question 5: System Performance Optimization Under Operational Constraints

Model	Accuracy (%)	Latency (ms)	Hardware / Operational Cost
Model A	88%	50 ms	Low
Model B (Selected)	92%	80 ms	Moderate
Model C	95%	120 ms	High

Balanced Trade-off Analysis

- Model A Evaluation:** Offers low latency (50 ms) and low cost, but sacrificing accuracy to 88% increases false negative risks in safety-critical applications.
- Model C Evaluation:** Provides 95% accuracy, but 120 ms latency violates real-time processing constraints (<10 FPS) and requires expensive infrastructure.
- Model B Evaluation:** Delivers the optimal balance: 92% accuracy guarantees strong detection reliability, 80 ms latency enables real-time frame rates (~12.5 FPS), and moderate hardware cost ensures economic scalability.

Key Recommendation / Verdict: Model B provides the best overall balance of accuracy, latency, and cost, making it the ideal selection for real-time deployment constraints.

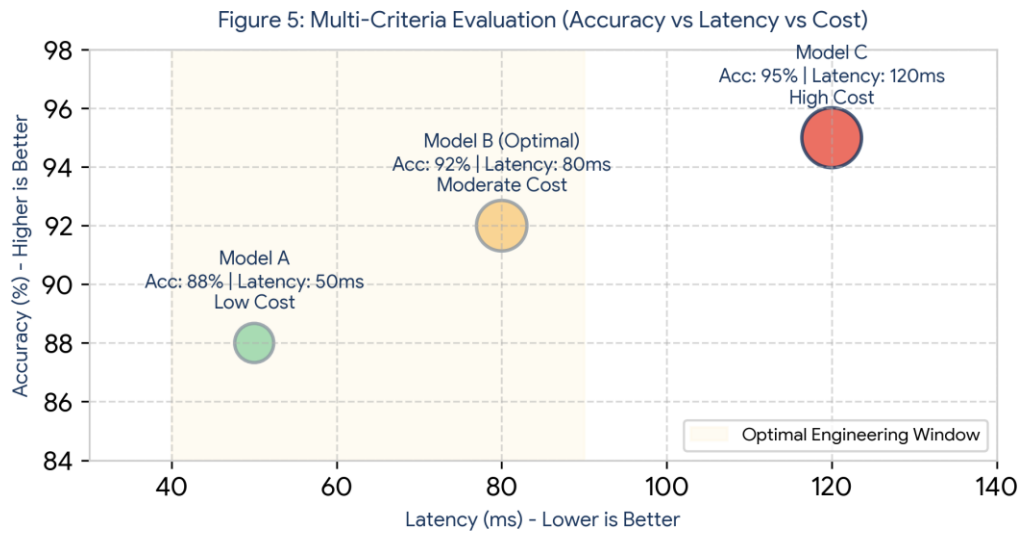


Figure 5: Multi-dimensional evaluation plot analyzing accuracy, latency, and deployment cost trade-offs.